

MULTIPLE CHOICE QUESTIONS: choose the best answer.

1. Mr Abebe was a diabetic patient who developed cellulitis due to *S.aureus*, which was found to be methicillin resistant on the antibiotic sensitivity testing. All the following antibiotics will be appropriate except:
A] Imipenem C] Vancomycin
B] Linezolid D] Teichoplanin
2. *Staphylococcus aureus* differs from *Streptococcus pyogenes* by:
A] Is catalase positive C] SA common cause of UTI
B] Is coagulase positive D] Causes endocarditis of prosthetic valve
3. Gram positive organism that most commonly cause of UTI among sexually active women:
A] *Staphylococcus aureus* C] *Enterococcus*
B] *Staphylococcus saprophyticus* D] *Staphylococcus epidermidis*
4. A 30-year-old female is on antibiotics with prolonged IV cannulation, has spike of fever, the likely cause is:
A] *Streptococcus agalactiae* C] Coagulase -ve staphylococcus
B] *E. coli* D] *Pseudomonas aeruginosa*
5. A chief prepares sandwiches for 10 people going for church picnic. Eight out of them develop severe gastroenteritis within 4-6 hrs of consumption of the sandwiches. It is likely that on investigations the cook is found to be carrier of:
A] *Salmonella typhi* C] *Vibrio cholerae*
B] *Staphylococcus aureus* D] *Entamoeba histolytica*
6. Factors responsible for the pathogenicity (virulence factor) of *Staphylococcus* can cause:
A] Teichoic acids C] Staphylokinase
B] Polysaccharide capsule (Microcapsule) D] Impetigo contagiosa
7. *Staphylococcus* can cause:
A] Pyogenic inflammation C] Furuncle
B] Post surgical wound infections D] Impetigo contagiosa
8. Quickest food poisoning (1-6) hour is:
A] *Staphylococcus* C] *Salmonella*
B] *B. cereus* D] *Vibrio cholera*
9. Toxic shock syndrome was commonly found in:
A] Tampon users C] Drug addicts
B] Diabetic septicemia D] None
10. All are capsulated Bacteria except:
A] *Step. Pneumonia* C] *Vibriocholera*
B] *Kleb. Pneumonia* D] *H. influenza*
11. Gram positive cocci are:
A] *Staphylococcus* C] *Pneumococcus*
B] *Streptococcus* D] All of the above
12. Ato Bekele comes from dinner and he complain of diarrhoea, vomiting after 4 hours of meal. Most likely causative agent:
A] *Stap aureus* C] *Streptococcus*
B] *V. cholerae* D] *E. coli*
13. Toxin of staphylococcus:
A] TSST C] Enterotoxins
B] Leucocidin D] All
14. Toxic shock syndrome is due to:
A] *Streptococcus pyogenes* C] *Strept albicans*
B] *Staphylococcus aureus* D] *E. durans*
15. Novobiocin susceptible staph is:
A] *S. aureus* C] *S. hemolyticus*
B] *S. epidermidis* D] *S. saprophyticus*
16. In which bacteria that immunity produced against its cell wall component cross react with myocardial muscle?
A] *S. aureus* C] *S. hemolyticus*
B] *St pneumonia* D] *St. pyogenes*

17. A patient admitted to St Paul's Hospital ICUs is on central venous line for the last one week; he is on ceftazidime and amikacin. After 7 days of antibiotics he develops a spike of fever and his blood culture is positive for gram positive cocci in chains, which are catalase negative. Following this, vancomycin was restarted but the culture remained positive for same organisms even after 2 weeks of therapy. The most likely organism causing infection is:
- A] Staph aureus C] Enterococcus faecalis
 B] Viridans streptococci D] Coagulase positive staphylococcus
18. Toxin involved in the streptococcal toxic shock syndrome is:
- A] Pyrogenic toxin C] Hemolysin
 B] Streptokinase D] Neurotoxin
19. True statement regarding Pneumococcus is:
- A] Virulence is due to polysaccharide capsule
 B] Capsule is protein in nature
 C] Antibodies against capsule are not protective
 D] Resistance to penicillin has not yet been reported
20. Lance field grouping of streptococci is done by using:
- A] M protein C] Group C carbohydrate antigen
 B] Group C peptidoglycan cell wall D] M antigen
21. A patient admitted to an ICU is on central venous line for the last one week. He is on Ceftazidime and amikacin. After 7 days of antibiotics he develops a spike of fever and his blood culture is positive for gram positive cocci in chains which are catalase - negative. Following this vancomycin was started but the culture remained positive for same organism even after 2 weeks of therapy. The most likely organism causing infection is:
- A] Staphylococci aureus C] Enterococcus faecalis
 B] Viridans streptococci D] Coagulase -ve staphylococcus
22. The sputum specimen of a 65-year-old male was cultured on a 5% blood agar. The culture showed the presence of a-haemolytic colonies. This organism is most likely:
- A] Gram positive cocci in short chains, catalase negative and bile resistant
B] Gram positive cocci in pairs, catalase negative and bile soluble
 C] Gram positive cocci in clusters, catalase positive and coagulase positive
 D] Gram negative coccobacilli, catalase positive and oxidase positive
23. Most common causative organism for lobar pneumonia is:
- A] Staphylococcus aureus C] Streptococcus pneumoniae
 B] Streptococcus pyogenes D] Haemophilus influenzae
24. The commonest organism causing cellulitis is:
- A] Streptococcus pyogenes C] Streptococcus viridans
 B] Streptococcus faecalis D] Microaerophilic streptococci
25. Risk of Pneumococcal meningitis is seen in:
- A] Post splenectomy patient
 B] Patient following cardiac surgery
 C] Patient undergone neurosurgical intervention
 D] Patient with hypoplasia of lung
26. C-Carbohydrate in beta hemolytic Streptococcus is important for
- A] Lancefield classification C] Toxin production
 B] Phagocytic inhibition D] Haemolysis
27. Drug of choice for the treatment of Streptococcus pneumonia is:
- A] Amoxicillin C] Chloramphenicol
B] Third generation Cephalosporins D] Staphylococci
28. Lymphangitis is caused by
- A] Staphylococcus C] Streptococci
 B] Pneumococci D] Neisseria
29. Which of the following factor is mainly responsible for virulence in Streptococcus:
- A] Carbohydrate C] Streptodornase
 B] Streptokinase D] M protein
30. Which toxin of streptococcus causes hemolysis:
- A] Streptolysin O C] Streptokinase
B] Streptolysin S D] Streptodornase
31. Streptococci causing dental caries:

- A] Streptococci equisimilis C] Streptococci pneumoniae
 B] Streptococci mutans D] Streptococci bovis
32. The following statements about meningococcal meningitis are true except:
 A] The source of infection is mainly clinical cases
 B] Chemoprophylaxis of close contacts of cases is recommended
 C] The disease is more common in dry and cold months of the year
 D] The vaccine is not effective in children below 2 years of age
33. Differentiation of *N. gonorrhea* and *N. meningitidis* can be done by:
 A] Glucose fermentation C] Maltose fermentation
 B] VP Reaction D] Indole test
34. The best site to obtain a swab in asymptomatic gonorrhea is:
 A] Endocervix C] Lateral vaginal wall
 B] Urethra D] Posterior fornix
35. The virulence factor of *Neisseria gonorrhoeae* includes all of the following except:
 A] Outer membrane proteins C] M Protein
 B] IgA Protease D] Antigenic variations
36. All are true about *Neisseria gonorrhea* except:
 A] Gram positive cocci
 C] Involves seminal vesicles and spreads to epididymis
 B] Causes stricture urethra
 D] Drug of choice is ceftriaxone
37. Which of the following is true regarding *Neisseria meningitidis* infection?
 A] It is the most common cause of meningitis in children
 B] In children less than 2 years the vaccine is not effective
 C] All strains are uniformly sensitive to sulfonamides
 D] In India sero type B is most common cause
38. The following statements are true regarding *Clostridium perfringens* except:
 A] It is the commonest cause of gas gangrene
 B] The principal toxin of *C. perfringens* is the alpha toxin
 C] It is normally present in human faeces
 D] Gas gangrene producing strains of *C. perfringens* produces heat resistant spores
39. A person has received complete immunization against tetanus 10 years ago. Now he presents with a clean wound without any lacerations from an injury sustained 2.5 hours ago. He should now be given:
 A] Full course of tetanus toxoid
 B] Human tet. globulin
 C] Single dose of tetanus toxoid
 D] Human tet. globulin and single dose of toxoid
40. True regarding pseudomembranous colitis are all except:
 A] It is caused by *Clostridium difficile*
 B] It is due to production of phospholipase A
 C] The organism is a normal commensal of gut
 D] It is treated by vancomycin
41. The Most common organism responsible for gas, gangrene is:
 A] *Clostridium-perfringens* C] *Clostridium tetani*
 B] *Clostridium-difficile* D] *Clostridium botulism*
42. Most common organism, responsible for pseudo membranous colitis is:
 A] *Clostridium difficile* C] *Clostridium bifermentans*
 B] *Clostridium botulism* D] *Clostridium histolyticum*
43. All of the following statements about Botulism are true except:
 A] Botulism is caused by endotoxin
 C] Constipation is seen
 B] Honey ingestion causes infant botulism
 D] Detection of antitoxin in the serum can aid in diagnosis
44. Swarming growth on culture is characteristic of which Gram-negative organism:
 A] *Clostridium welchi* C] *Bacillus cereus*
 B] *Clostridium tetani* D] *Proteus mirabilis*
45. An 18-year-old male presented with acute onset descending paralysis of 3 days duration. There is also a history of blurring of vision for the same duration. On examination, the patient has quadriparesis with areflexia. Both the pupils are non-reactive. The most probable diagnosis is:

- A] Poliomyelitis
 B] Botulism
- C] Diphtheria
 D] Porphyria
46. A patient of Acute lymphocytic leukemia with fever and neutropenia develops diarrhoea after administration of amoxicillin therapy, which of the following organism is most likely to be the causative agent?
 A] Salmonella typhi
 B] Clostridium difficile
 C] Clostridium perfringens
 D] Shigella flexneri
47. A 10-year-old boy following a road traffic accident presents to the casualty with contaminated wound over the left leg. He has received his complete primary immunization before preschool age and received a booster of DT at school entry age. All of the following can be done except:
 A] Injection of TT
 B] Broad spectrum antibiotics
 C] Injection of human antiserum
 D] Wound debridement and cleaning spores
48. The most effective way of preventing tetanus is:
 A] Surgical debridement and toilet
 B] Hyperbaric oxygen
 C] Antibiotics
 D] Tetanus toxoid
49. Gastro-intestinal enteritis necroticans caused by:
 A] Cl. difficile
 B] Cl. perfringens
 C] Botulinum
 D] C. jejuni
50. True about clostridium tetani:
 A] It is gram +ve
 B] Drum stick appearance
 C] Grows in aerobic environment
 D] It is gram -ve
51. True statement regarding diphtheria is:
 A] It can be diagnosed by demonstration of antibodies by ELISA
 B] Treatment of contacts is not indicated
 C] Immunization prevents carriers state
 D] Iron has critical value in the production of toxin
52. A man, after skinning a dead animal, developed a pustule on his hand. A smear prepared from the lesion showed the presence of Gram-positive bacilli in long chains which were positive for McFadyean's reaction. The most likely aetiological agent is:
 A] Clostridium tetani
 B] Listeria monocytogenes
 C] Bacillus anthracis
 D] Actinomyces sp
53. All of the following are true about anthrax except:
 A] Plasmid is responsible for toxin production
 B] Capsular polypeptide aids virulence by inhibiting phagocytosis
 C] Cutaneous anthrax generally resolves spontaneously
 D] Toxin is a complex of two fractions
54. True regarding anthrax is all except:
 A] Caused by insect bite
 B] Cutaneous type is rare now days
 C] Caused by rubbing of skin
 D] Pulmonary infection occurs by inhalation
55. A patient present with vomiting, he had eaten rice 6 hours before. The most probable cause is:
 A] Bacillus cereus
 B] Staph. aureus
 C] Cl. difficile
 D] All
56. A 30-year-old woman with a bad obstetric history presents with fever. The blood culture from the patient grows gram-positive small to medium coccobacilli that are pleomorphic, occurring in short chains. Direct wet mount from the culture shows tumbling motility. The most likely organism is:
 A] Listeria monocytogenes
 B] Corynebacterium sp.
 C] Enterococcus sp
 D] Erysipelothrix rhusiopathiae
57. All the following are true about Listeria except:
 A] Transmitted by contaminated milk
 B] Gram (-)ve bacteria
 C] Causes abortion in pregnancy
 D] Causes meningitis in neonates
58. A major step in the pathogenesis of listeriosis is:
 A] The formation of antigen-antibody complex with resultant complement activation and tissue damage

- B] The antiphagocytic activity of the *L. monocytogenes* capsule
 C] The release of hyaluronidase by *L. monocytogenes*, which contributes to its dissemination from local sites
D] The survival and multiplication of *L. monocytogenes* within mononuclear phagocytes and host epithelial cells
59. In patient with *Listeria meningitidis* who is allergic penicillin the treatment of choice is :
 A] Vancomycin **C] Trimethoprim - sulphamethoxazole**
 B] Gentamycin D] Ceftriaxone
60. The medium used for *Mycobacterium tuberculosis* is :
 a) Sabouraud's medium c) Pick's medium
b) L J medium d) NIH medium
61. Mycobacterial lesion that consists of central area of giant cells containing tubercle bacilli surrounded by a zone of epithelioid cells:
a) Granulomatous lesions c) Exudative lesion
 b) pyrogenic lesion d) Cellulitis
62. In TB, immunity is provided by:
a) CD 4+ c) IgG
 b) CD 8+ d) IgM
63. With reference to infection with *Escherichia coli* the following are true except:
 a) Enteraggregative *E. coli* is associated with persistent diarrhoea
 b) Enterohemorrhagic *E. coli* cause hemolytic uremic syndrome
c) Enteroinvasive *E. coli* produces a disease similar to salmonellosis
 d) Enterotoxigenic *E. coli* is a common cause of traveller's diarrhoea
64. A 20-year-old man presented with hemorrhagic colitis. The stool sample grew *Escherichia coli* in pure culture. The following serotype of *E. coli* is likely to be the causative agent:
a) O 157:H7 c) O 107:H7
 b) O 159:H7 d) O 55:H7
65. *Shigella* can be differentiated from *E. coli* by all of the following features except:
 a) *Shigella* does not produce gas from glucose
 b) *Shigella* does not ferment lactose
c) *Shigella* does not ferment mannitol
 d) *Shigella* has no flagella and is non motile
66. Which of the following is a true statement regarding Enterobacteriaceae:
 a) Motility is by polar flagellum
b) Glucose is not fermented by all members of the family
 c) All members are oxidase positive
 d) Nitrate reduction negative
67. Most common organism causing UTI:
a) *E. coli* c) *Klebsiella*
 b) *Proteus* d) *Staphylococci*
68. Which of the following produces enterotoxin:
a) *Sh. dysenteriae* c) *Sh. flexneri*
 b) *Sh. sonnei* d) *Sh. boydi*
69. *Salmonella typhi* is the causative agent of typhoid fever. The infective dose of *S. typhi* is:
 a) One bacillus **c) 10² - 10⁵ bacilli**
 b) 10⁸ - 10¹⁰ bacilli d) 1 - 10 bacilli
70. A 20-year-old male had pain in abdomen and mild fever followed by gastroenteritis. The stool examination showed presence of pus cells and RBC's on microscopy. The most likely etiological agent is
a) Enteroinvasive *E. coli* c) Enterotoxigenic *E. coli*
 b) Enteropathogenic *E. coli* d) Enteraggregative *E. coli*
71. All are true about *Shigella* except :
a) Large dose is required for infection
 b) Associated with hemolytic uremic syndrome
 c) Causes bloody diarrhoea with mucus
 d) Gut pathology is due to toxin

72. A person returns to Addis from Shashamane, after 2 days he has diarrhoea. Stool examination shows RBCs in stool. The likely organism causing is:
- a) Enteropathogenic *E. coli*
 - b) Enterotoxigenic *E. coli*
 - c) *Salmonella typhi*
 - d) *Shigella dysenteriae*
73. Toxin is mediated by C-AMP except:
- a) *V. cholera* 01
 - b) Heat stable *E. coli* toxin
 - c) Heat labile *E. coli* toxin
 - d) *V. cholera* 0137
74. Traveller's diarrhoea, is caused by:
- a) *Shigella*
 - b) *E. coli*
 - c) *E. histolytica*
 - d) Giardiasis
75. Enterobacteriaceae all except:
- a) *Pseudomonas*
 - b) *Klebsiella*
 - c) *V. cholera*
 - d) *Proteus*
 - e) *E. Coli*
76. True about salmonella gastroenteritis is/are:
- a) Mainly diagnosed by serological tests
 - b) Blood & mucous are present in stool
 - c) Caused by animal products
 - d) Symptoms appear between 4 - 48 hours
77. True about typhoid:
- a) It is caused by *S. typhi*
 - b) Water can transmit the disease
 - c) Type 21 a is an oral vaccine
 - d) Chronic carrier is called when transmitted up to 6 month
78. Which of these are true about *E. coli* :
- a) The L.T., (labile toxin), in ETEC acts via CAMP
 - b) UTI causing *E. coli* attaches through pili
 - c) The ST (Stable toxin) of ETEC is responsible for causing hemolytic-uremic syndrome
 - d) EIEC invasiveness is under plasmid control
79. Antibiotics of choice for the treatment Enterocolitis caused by *Salmonella typhi*:
- a) Ciprofloxacin
 - b) Gentamycin
 - c) Can resolve without antibiotics treatment
 - d) fluid-electrolyte replacement
80. In the small intestine, cholera toxin acts by:
- a) ADP ribosylation of G regulatory protein
 - b) Inhibition of adenyl cyclase
 - c) Activation of GTPase
 - d) Active absorption of NaCl
81. Antibiotic treatment of choice for treating cholera in an adult is a single dose of:
- a) Tetracycline
 - b) Cotrimoxazole
 - c) Doxycycline
 - d) Furazolidone
82. Cholera toxin:
- a) Increases the levels of intracellular cyclic GMP
 - b) Acts through the receptor for opiates
 - c) Causes continued activation of adenylate cyclase
 - d) Inhibits the enzyme phosphodiesterase
83. True about *V. cholerae* is:
- a) One attack of *V. cholerae* gives life-long immunity
 - b) Affects adults and children with equal propensity in non epidemic regions
 - c) In between epidemics, carrier states maintain the organism
 - d) Pathogenicity of 0-139 vibrio is due to O antigen
84. Cholera transmission by:
- a) Food transmits
 - b) Vaccination gives 90% efficiency
 - c) Healthy carrier
 - d) Chlorination is not effective
85. True about cholera vibrios is:
- a) Can tolerate wide range of alkaline pH
 - b) Non motile bacilli
 - c) Can't be grown in media
 - d) NaCl stimulates growth

86. Which one of the following bacteria are pigmented?
 a) Staph. aureus c) Serratia
 b) Pseudomonas d) E. coli
87. An organism grown on agar shows green colored colonies; likely organism is:
 a) Staphylococcus c) Pseudomonas
 b) E. Coli d) Peptostreptococcus
88. The drug of choice for chemoprophylaxis in contacts of a patient of pneumonic plague is :
 a) Penicillin c) Erythromycin
 b) Rifampicin d) Tetracycline
89. True about Y. pestis:
 a) Gram +ve c) Motile e) It is coccobacilli
 b) Gram -ve d) Non-motile
90. A veterinary doctor had pyrexia of unknown origin. His blood culture in special laboratory media was positive for gram negative short bacilli which was oxidase positive. Which one of the following is the likely organism grown in culture?
 a) Pasturella spp. c) Bartonella spp.
 b) Francisella spp. d) Brucella spp.
91. The following are true for Bordetella pertussis except:
 a) It is a strict human pathogen
 b) It can be cultured from the patient during catarrhal stage
 c) It leads to invasion of the respiratory mucosa
 d) Infection can be prevented by a acellular vaccine
92. Malta fever is caused by:
 a. Legionella c. Brucella melitensis
 b. Borrelia burgdorferi d. Pseudomonas
93. Brucella is transmitted by all the following means except:
 a) Through placenta of animals c) Aerosol
 b) Person to person transmission d) Eating uncooked food
94. All the following are true about H. influenzae except:
 a) It can be part of normal flora of some persons
 b) The serotyping is based on bacterial outer membrane protein
 c) It requires Haemin and NAD for growth in culture medium
 d) Type b is responsible for invasive disease
95. A 2 years old child is brought to the emergency with history of fever and vomiting. On examination he has neck rigidity. CSF examination shows polymorphs more than 200/ μ l; protein 100mg/dl and glucose 10mg/dl. the Grams stain shows the presence of Gram negative coccobacilli. The culture shows of bacteria only on chocolate. Agar and not on blood agar. The causative agent is:
 a) Neisseria meningitides c) Branhamella catarrhalis
 b) Haemophilus influenzae d) Legionella pneumophila
96. Regarding Brucellosis, all of these are true except:
 a) Man to man transmission
 b) A zoonosis
 c) Blood cultures used in diagnosis
 d) Transmitted via animal products
97. True about helicobacter pylori is :
 a) Culture and gram staining of biopsy is the gold standard investigation
 b) Controlled urea breath is negative with massive infection
 c) Anti urease antibody are produced only by invasive strains
 d) Ureas activity provides protective environment to the bacilli
98. Helicobacter pylori is not associated with:
 a) Gastrointestinal lymphoma c) Gastric leiomyoma
 b) Gastric cancer d) Peptic ulcer
99. The most sensitive test for H. pylori is:
 a) Breath test c) Culture of biopsy
 b) Rapid urease test d) Microscopy of biopsy

MATCHING: Select the best answer and match (1 point each)

A

- 1. Genital infections (B)
- 2. Toxic shock syndrom toxins (TSST) (D)
- 3. Streptolysin O (E)
- 4. Swarming colonies on medium (G)
- 5. Giant cell formation (A)
- 6. Gram positive intracellular bacteria (G)
- 7. Enzyme that assist spreading of streptococcus (F)

B

- A. Mycobacterium tuberculosis
- B. Staphylococcus saprophyticus
- C. Oxygen stable
- D. Staphylococcus aureus
- E. Oxygen labile
- F. Staphylokinase
- G. Listeria monocytogenes
- H. Streptokinase
- I. Vibrio
- J. Proteus

FILL-IN THE SPACE: Place your answer in the blank (1 point each).

1. _____ and _____ are enzymes used in the spreading of staphylococcus and streptococcus, respectively (Staphylokinase and Streptococcus).
2. Wild rodents are the natural hosts of _____ (Y. pestis)
3. _____ is a treat for biological war (Bacillus anthrax)
4. _____ is a bacteria that cause cancer in human (H. pylori)
5. _____ causes antibiotic associated diarrhea (C. difficile)

SHORT ANSWER: Answer each in your booklet

1. List immunological diseases caused after the infection of Streptococcus Pyogens ? (2pts)
(rheumatic fever and acute glomerulonephritis)
2. List at least 3 strains of Shigella (3pts)
(S. dysenteriae, S. boydii, S. flexneri and S. sonnei)
3. Mechanism of action of Botulinum toxin? (2pts)

The toxin is proteases that cleave proteins involved in acetylcholine release, From the site of entry into the body (gut and wound), the toxin travels through the blood and lymphatic system and then becomes fixed to cranial and peripheral nerves thereby inhibiting the release of acetylcholine at the synaptic sites resulting flaccid muscular paralysis.

4. Mechanism of cholera toxins in causing diarrhea? (2pts)

Cholera toxin is an exotoxin that activates the adenylate cyclase enzyme in cells of the intestinal mucosa leading to increased levels of intracellular cAMP, and the secretion of H₂O, Na⁺, K⁺, Cl⁻, and HCO₃⁻ into the lumen of the small intestine.